

SWIGGY SALES AND RESTAURANT ANALYTICS

DATASET OVERVIEW

The dataset contains restaurant menu listings from Swiggy, where each record represents a menu item offered by a restaurant. It includes information about restaurants, locations, food categories, dish names, prices, and customer ratings. The dataset is designed to support analytical queries for understanding restaurant performance, pricing trends, cuisine distribution, and customer preferences.

Column	Description
Restaurant_Name	Name of the restaurant
Location	Area/locality where the restaurant is located
Category	Food category or cuisine of the dish
Dish_Name	Name of the menu item
Price_INR	Price of the dish (in INR)
Rating	Customer rating of the dish

Dataset Characteristics

- Each row represents a **restaurant–dish combination**.
- The same restaurant can offer multiple dishes across different categories.
- The same dish may be available at multiple restaurants.
- The dataset includes menu pricing and customer ratings across multiple restaurants and locations.
- The data is suitable for analyzing restaurant performance, pricing patterns, cuisine popularity, and location-based trends using SQL.

BUSINESS REQUIREMENTS

Project Objective

The objective of this project is to design a structured SQL database using a Star Schema and perform analytical queries on Swiggy restaurant menu data to generate business insights.

The analysis focuses on restaurant performance, menu pricing, cuisine distribution, customer ratings, and location-based trends to support data-driven decision making.

Business Requirements

The business requires insights into the following areas:

Restaurant Performance

- Identify the highest-rated restaurants.
- Determine restaurants offering the largest number of menu items.
- Analyze the average menu price across restaurants.

Pricing Analysis

- Calculate the overall average dish price.
- Identify the most expensive dishes available on the platform.
- Analyze the distribution of menu items across different price ranges.

Category Analysis

- Identify food categories with the largest number of menu items.
- Compare the average dish price across different food categories.
- Determine the highest-rated food categories.

Dish Analysis

- Identify the highest-rated dishes.
- Determine the average rating of dishes.
- Analyze the average price of different dishes.

Location Analysis

- Identify locations with the highest number of restaurants.
- Determine locations with the largest number of menu items.
- Compare average dish prices across different locations.

Rating Analysis

- Calculate the overall average customer rating.
- Analyze the distribution of menu items across different rating levels.

Data Cleaning & Validation

Before performing the analysis, the dataset was validated to ensure consistency and reliability.

The following checks were performed:

- Checked for missing (NULL) values across all columns.
- Identified blank or inconsistent text values.
- Detected duplicate records using business-critical attributes.
- Removed duplicate records while retaining a single valid observation.
- Standardized the cleaned dataset for dimensional modelling and SQL analysis.

Expected Outcomes

The analysis is expected to provide insights that help:

- Evaluate restaurant performance using customer ratings and menu offerings.
- Understand pricing trends across restaurants and food categories.
- Compare restaurant and menu distribution across different locations.
- Identify highly rated dishes and popular food categories.
- Generate meaningful business insights using SQL for reporting and decision-making.

DATABASE DESIGN

Star Schema Design

To support efficient querying and simplify analytical reporting, the cleaned dataset was organized using a **Star Schema**. This dimensional model separates descriptive information into dimension tables while storing measurable attributes in a central fact table.

The Star Schema reduces data redundancy, improves query performance, and provides a scalable structure for business analysis.

Database Schema

The database consists of **five dimension tables** and **one fact table**.

Dimension Tables

Table	Description
dim_date	Stores date-related attributes such as day, month, quarter, and year.
dim_location	Stores location information including state, city, and locality.
dim_restaurant	Stores unique restaurant names.
dim_category	Stores food categories or cuisines.
dim_dish	Stores unique dish names.

Fact Table

Table	Measures
fact_swiggy_orders	Price (INR), Rating, and foreign keys referencing all dimension tables.

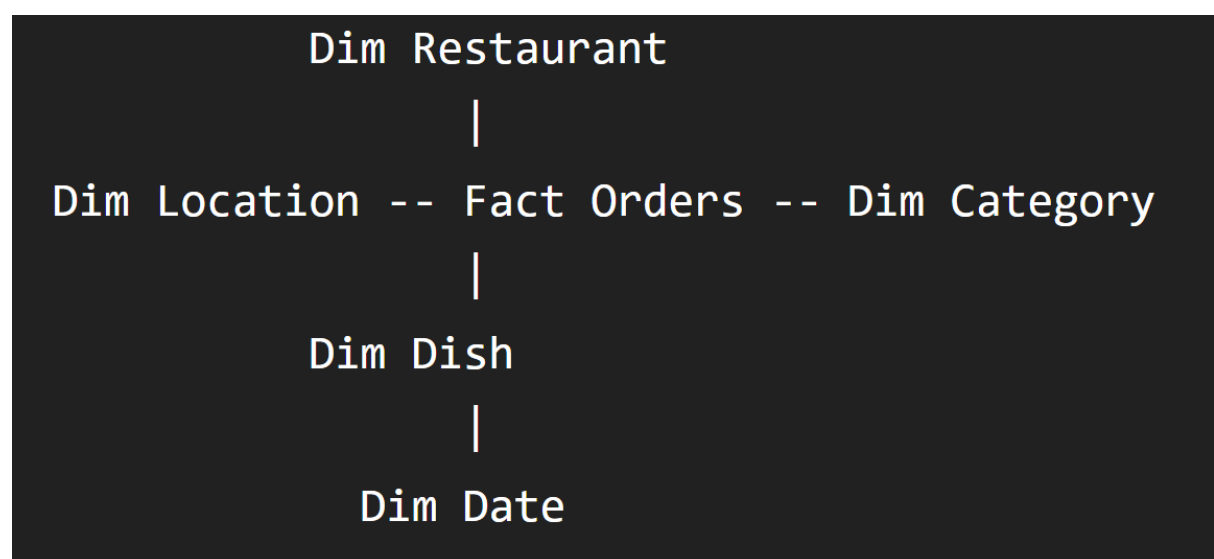
Star Schema Relationships

The **fact_swiggy_orders** table acts as the central table and connects to each dimension table through foreign keys.

Relationship Summary:

- `date_id` → `dim_date`
- `location_id` → `dim_location`
- `restaurant_id` → `dim_restaurant`
- `category_id` → `dim_category`
- `dish_id` → `dim_dish`

ERD DIAGRAM: STAR SCHEMA



Benefits of the Star Schema

- Reduced data redundancy through normalization of descriptive attributes.
- Improved SQL query performance using dimension-based joins.
- Simplified analytical reporting for restaurant, dish, category, and location analysis.
- Better scalability for adding future analytical metrics and dimensions.

SQL ANALYSIS AND BUSINESS INSIGHTS

Overview

After designing the Star Schema, SQL queries were developed to analyze restaurant performance, pricing trends, cuisine distribution, customer ratings, and location-wise patterns. The queries utilized joins, aggregate functions, Common Table Expressions (CTEs), window functions, and date functions to generate meaningful business insights.

Key SQL Analysis

Analysis Area	Business Objective	SQL Concepts Used
Restaurant Performance	Identify highest-rated restaurants and menu diversity	JOIN, GROUP BY, AVG(), COUNT()
Pricing Analysis	Analyze average prices and price distribution	AVG(), CASE, GROUP BY
Category Analysis	Compare cuisine popularity and pricing	JOIN, GROUP BY, ORDER BY
Dish Analysis	Identify highest-rated and premium dishes	JOIN, AVG(), MAX()
Location Analysis	Compare restaurants and pricing across locations	JOIN, GROUP BY
Rating Analysis	Analyze rating distribution and average ratings	AVG(), COUNT(), GROUP BY

KPI Development

Once the schema is built, compute core performance indicators:

Basic KPIs

- Total Orders
- Total Revenue (INR Million)
- Average Dish Price
- Average Rating

Deep-Dive Business Analysis

Date-Based Analysis

- Monthly order trends
- Quarterly order trends
- Year-wise growth
- Day-of-week patterns

Location-Based Analysis

- Top 10 cities by order volume
- Revenue contribution by states

Food Performance

- Top 10 restaurants by orders
- Top categories (Indian, Chinese, etc.)
- Most ordered dishes
- Cuisine performance → Orders + Avg Rating

Customer Spending Insights

Buckets of customer spend:

- Under 100
- 100–199
- 200–299
- 300–499
- 500+

With total order distribution across these ranges

Sample Business Insights

The SQL analysis generated several meaningful insights, including:

- Restaurants with consistently high customer ratings were identified.
- Significant price variation was observed across restaurants and food categories.
- Certain locations contained a higher concentration of restaurants and menu items.
- Premium-priced dishes were concentrated within specific cuisines.
- Menu item distribution varied considerably across different food categories.

Project Outcome

A total of **20+ SQL queries** were developed to answer business questions and generate actionable insights from the dataset. The analysis demonstrates how SQL can be used to transform raw restaurant menu data into structured information for business reporting and decision-making.

Top 10 Highest Rated Restaurants

```

450
451 -- Top 10 Highest Rated Restaurants
452
453 SELECT
454     r.restaurant_name,
455     COUNT(*) AS Total_Orders,
456     ROUND(AVG(f.rating), 2) AS Average_Rating
457 FROM fact_swiggy_orders f
458 JOIN dim_restaurant r
459     ON r.restaurant_id = f.restaurant_id
460 GROUP BY
461     r.restaurant_name
462 ORDER BY
463     Average_Rating DESC,
464     Total_Orders DESC
465 LIMIT 10;

```

	restaurant_name character varying (200)	total_orders bigint	average_rating numeric
1	Sakana	95	4.82
2	Vijay Dairy	52	4.75
3	Jagannath Mandir Arna Prasad	8	4.75
4	Radhey Lal's Parampara Sweets	84	4.74
5	I Deli Cafe	47	4.74
6	Yadav Doodh Dairy	70	4.73
7	Tossin Pizza	158	4.71
8	Perambur Sri Srinivasa Sweets & Sna...	20	4.71
9	Wooden Plate	53	4.69
10	Natural Ice Cream	87	4.68

Total Orders Month Wise

```

247 -- Total Orders month wise
248 SELECT
249     d.year,
250     d.month,
251     d.month_name,
252     COUNT(*) AS Total_Monthly_Orders
253 FROM fact_swiggy_orders f
254 JOIN dim_date d
255     ON f.date_id = d.date_id
256 GROUP BY
257     d.year,
258     d.month,
259     d.month_name
260 ORDER BY Total_Monthly_Orders DESC;
261

```

	year integer	month integer	month_name character varying (20)	total_monthly_orders bigint
1	2025	1	January	25393
2	2025	8	August	25227
3	2025	5	May	25188
4	2025	7	July	24936
5	2025	4	April	24584
6	2025	3	March	24400
7	2025	6	June	24383
8	2025	2	February	23292

Total Orders Month Wise

```

431 -- Order Value Distribution (Total Orders by Price Range)
432 SELECT
433     price_range,
434     COUNT(*) AS total_orders
435 FROM (
436     SELECT
437         CASE
438             WHEN price_inr < 100 THEN 'Under 100'
439             WHEN price_inr BETWEEN 100 AND 199 THEN '100 - 199'
440             WHEN price_inr BETWEEN 200 AND 299 THEN '200 - 299'
441             WHEN price_inr BETWEEN 300 AND 499 THEN '300 - 499'
442             ELSE '500+'
443         END AS price_range
444     FROM fact_swiggy_orders
445 ) t
446 GROUP BY
447     price_range
448 ORDER BY
449     total_orders DESC;
450

```

	price_range text	total_orders bigint
1	100 - 199	56190
2	200 - 299	54567
3	300 - 499	43758
4	Under 100	26796
5	500+	16092

Top 10 Cities by Total Revenue

```

330 -- Top 10 Cities by Total Revenue
331 SELECT
332     l.city,
333     CONCAT(ROUND(SUM(price_inr) / 1000000, 2), ' Million Rupees') AS Total_City_Revenue
334 FROM fact_swiggy_orders f
335 JOIN dim_location l
336     ON l.location_id = f.location_id
337 GROUP BY
338     l.city
339 ORDER BY
340     Total_City_Revenue DESC
341 LIMIT 10;
342

```

	city character varying (100)	total_city_revenue text
1	Bengaluru	5.46 Million Rupees
2	Lucknow	3.12 Million Rupees
3	Mumbai	3.02 Million Rupees
4	Hyderabad	3.02 Million Rupees
5	New Delhi	2.83 Million Rupees
6	Ahmedabad	2.82 Million Rupees
7	Chandigarh	2.80 Million Rupees
8	Kolkata	2.66 Million Rupees
9	Chennai	2.64 Million Rupees
10	Jaipur	2.50 Million Rupees

Revenue Contribution by States

```

343 -- Revenue Contribution by States
344 SELECT
345     l.state,
346     CONCAT(ROUND(SUM(price_inr) / 1000000, 2), ' Million Rupees') AS Total_State_Revenue
347 FROM fact_swiggy_orders f
348 JOIN dim_location l
349     ON l.location_id = f.location_id
350 GROUP BY
351     l.state
352 ORDER BY
353     Total_State_Revenue DESC;
354

```

	state character varying (100)	total_state_revenue text
1	Karnataka	5.46 Million Rupees
2	Uttar Pradesh	3.12 Million Rupees
3	Telangana	3.02 Million Rupees
4	Maharashtra	3.02 Million Rupees
5	Delhi	2.83 Million Rupees
6	Gujarat	2.82 Million Rupees
7	Punjab	2.80 Million Rupees
8	West Bengal	2.66 Million Rupees
9	Tamil Nadu	2.64 Million Rupees
10	Rajasthan	2.50 Million Rupees

CONCLUSION & KEY LEARNINGS

Project Summary

This project demonstrates the complete SQL analytics workflow, starting from data cleaning and Star Schema design to business-oriented analysis using SQL. By transforming raw restaurant menu data into a structured relational database, meaningful insights were generated regarding restaurant performance, pricing trends, cuisine distribution, customer ratings, and location-based patterns.

The project highlights how SQL can be used to organize data efficiently and support business decision-making through structured analytical queries.

Key Learnings

During this project, the following concepts were implemented and strengthened:

- Designed and implemented a **Star Schema** consisting of five dimension tables and one fact table.
- Performed data cleaning, validation, and duplicate removal to improve data quality.
- Applied SQL concepts including **JOINS, Aggregate Functions, CASE statements, Common Table Expressions (CTEs), Window Functions, GROUP BY, ORDER BY, and Date Functions**.
- Generated business insights by analyzing restaurant performance, pricing patterns, menu diversity, customer ratings, and location-based trends.
- Improved understanding of dimensional modelling and SQL-based analytical reporting.